

Tasks	Responsible	Status	Planned Start Date	Start Date	End Date	Planned End Date	Deliverable	Notes
Module 1: Project Setup & Development Environment		In Progress	27-Feb-2026			05-Mar-2026		
Initialize FastAPI project structure (app/, api/, agent/, security/, database/, models/ directories)		Not Started	27-Feb-2026			28-Feb-2026	Project skeleton with folder structure	Follow structure from Blueprint Section 4
Create requirements.txt with all dependencies (langchain, fastapi, aiodebc, hdbcli, ldap3, redis, sqlparse, pyjwt)		Not Started	27-Feb-2026			27-Feb-2026	requirements.txt	Pin all dependency versions for reproducibility
Set up environment-based configuration module (app/config.py) with Pydantic Settings		Not Started	28-Feb-2026			28-Feb-2026	config.py with all env vars	DB hosts, Redis URL, AD server, LLM API keys, JWT secret
Configure Docker Compose for local development (FastAPI, Redis, PostgreSQL for checkpointer)		Not Started	28-Feb-2026			01-Mar-2026	docker-compose.yml	Exclude business DBs — they are existing systems
Set up logging framework with structured logging (structlog)		Not Started	01-Mar-2026			01-Mar-2026	Logging configuration	Correlation IDs for request tracing across middleware
Create .env.example with all required environment variables documented		Not Started	01-Mar-2026			01-Mar-2026	.env.example	Include comments for each variable group
Set up Git repository structure, .gitignore, and branch strategy		Not Started	27-Feb-2026			27-Feb-2026	Repository initialized	
Configure pre-commit hooks (black, ruff, mypy) for code quality		Not Started	02-Mar-2026			02-Mar-2026	Pre-commit config	Enforce type hints and consistent formatting
Create FastAPI main.py with CORS configuration and lifespan events		Not Started	02-Mar-2026			03-Mar-2026	main.py with app startup/shutdown	Lifespan: init DB pools, Redis, agent on startup; close on shutdown
Implement health check endpoint (GET /api/v1/health) with per-database connectivity status		Not Started	03-Mar-2026			04-Mar-2026	Health endpoint	Blueprint Section 4 — include DB-level health checks
Module 2: Authentication & Identity Layer (Active Directory)		Not Started	06-Mar-2026			19-Mar-2026		
Implement Active Directory LDAP client (app/security/ad_client.py) using ldap3		Not Started	06-Mar-2026			08-Mar-2026	AD client module	NTLM auth, extract employeeID, department, email, displayName, memberOf
Implement parallel role resolution from Pharma CRM database (MS SQL)		Not Started	08-Mar-2026			09-Mar-2026	Pharma role resolver	Query pharma_users table by ad_username
Implement parallel role resolution from SAP HANA database		Not Started	08-Mar-2026			09-Mar-2026	SAP role resolver	Query sap_user_roles table by ad_user_id
Implement parallel role resolution from Third-Party database (MS SQL)		Not Started	08-Mar-2026			09-Mar-2026	Third-Party role resolver	Query tp_access_control table by ad_email
Build CompositeRoleProfile dataclass and parallel assembly logic (asyncio.gather)		Not Started	09-Mar-2026			11-Mar-2026	CompositeRoleProfile with assembly	Handle DB failures gracefully — down DB ≠ 'no access'
Implement role profile caching in Redis (TTL: 15 min)		Not Started	11-Mar-2026			12-Mar-2026	Role cache module (app/cache/role_cache.py)	Avoid re-querying all 3 DBs on every request
Implement JWT issuance with composite_role_ref, ad_user_id, employee_id, department		Not Started	12-Mar-2026			13-Mar-2026	JWT token generation	Use PyJWT; encode pointer to cached role profile
Implement JWT validation middleware for FastAPI (app/middleware/auth.py)		Not Started	13-Mar-2026			14-Mar-2026	Auth middleware	Validate token, hydrate SecurityContext from Redis cache + JWT claims
Build POST /api/v1/auth/login endpoint with full AD auth + role assembly flow		Not Started	14-Mar-2026			16-Mar-2026	Login endpoint	Blueprint Section 2 — full authentication flow
Implement rate limiting middleware (Redis-backed, per-user rolling window)		Not Started	16-Mar-2026			17-Mar-2026	Rate limiter (app/middleware/rate_limit.py)	Prevent abuse; configurable limits per user
Write integration tests for the complete authentication flow		Not Started	17-Mar-2026			19-Mar-2026	Auth test suite	Mock AD, mock DB role queries, verify JWT contents
Module 3: Multi-Database Connection Layer		Not Started	13-Mar-2026			27-Mar-2026		
Implement Pharma CRM connection pool (MS SQL via aiodebc)		Not Started	13-Mar-2026			15-Mar-2026	pharma_pool.py	Async pool, minsize=2, maxsize=10, ODBC Driver 18
Implement Third-Party database connection pool (MS SQL via aiodebc)		Not Started	13-Mar-2026			15-Mar-2026	thirdparty_pool.py	Same async pattern as Pharma pool
Implement SAP HANA connection pool wrapper (hdbcli with ThreadPoolExecutor for async)		Not Started	15-Mar-2026			18-Mar-2026	sap_pool.py (HanaConnectionPool class)	Blueprint Section 6 — hdbcli is synchronous, wrap with executor
Build DatabaseConnectionManager with initialize(), shutdown(), and per-DB health checks		Not Started	18-Mar-2026			20-Mar-2026	connections.py	Centralized pool lifecycle management
Integrate connection pools with FastAPI lifespan events (startup/shutdown)		Not Started	20-Mar-2026			21-Mar-2026	Updated main.py lifespan	Initialize all 3 pools on startup, close gracefully on shutdown
Implement per-database health check functions with timeout handling		Not Started	21-Mar-2026			22-Mar-2026	health.py	SELECT 1 for MS SQL, SELECT 1 FROM DUMMY for HANA; 5s timeout
Define Pharma CRM schema metadata (tables, columns, descriptions, scope_columns)		Not Started	22-Mar-2026			24-Mar-2026	Schema definitions for Pharma	Blueprint Section 6 — products, sales_orders, customers, etc.
Define SAP HANA schema metadata (EKKO, EKPO, BKPf, BSEG, MARA, etc.)		Not Started	24-Mar-2026			26-Mar-2026	Schema definitions for SAP	Uppercase double-quoted identifiers; include SAP SQL addendum

Define Third-Party schema metadata (vendor_ratings, logistics_shipments, etc.)		Not Started	24-Mar-2026			26-Mar-2026	Schema definitions for Third-Party	T-SQL dialect, scope_column per table
Write connection pool integration tests (verify connectivity, pool behavior, error handling)		Not Started	26-Mar-2026			27-Mar-2026	DB connection test suite	Test pool exhaustion, reconnection, timeout handling
Module 4: LangChain 1.0 Agent & Middleware Stack		Not Started	20-Mar-2026			10-Apr-2026		
Set up LLM provider abstraction (app/models/llm_provider.py) with init_chat_model		Not Started	20-Mar-2026			21-Mar-2026	LLM provider module	GPT-4.1 primary; configure fallback to GPT-4o
Define AtcoGenieAgentState schema (security_context, last_intent, last_target_system, resolved_entities)		Not Started	21-Mar-2026			22-Mar-2026	Agent state schema	Extends AgentState from LangChain 1.0
Implement SecurityContextMiddleware (before_model hook — inject per-system context)		Not Started	22-Mar-2026			25-Mar-2026	security.py middleware	Blueprint Section 9 — inject and validate security context on every turn
Implement DynamicToolFilterMiddleware (wrap_model_call — per-system RBAC tool filtering)		Not Started	25-Mar-2026			28-Mar-2026	tool_filter.py middleware	Remove tools for unauthorized systems based on accessible_systems
Configure SummarizationMiddleware (built-in LangChain 1.0, gpt-4o-mini, 4000 token threshold)		Not Started	28-Mar-2026			29-Mar-2026	summarization.py config	Blueprint Section 11 — automatic context compression
Implement UsageTrackingMiddleware (after_model hook — log tokens, latency, target system)		Not Started	29-Mar-2026			30-Mar-2026	usage.py middleware	Integrate with LangSmith/Langfuse callbacks
Build the complete system prompt (ATCOGENIE_SYSTEM_PROMPT) with routing examples and rules		Not Started	30-Mar-2026			02-Apr-2026	System prompt definition	Blueprint Section 5.1 — 3 systems, routing examples, disambiguation rules
Build SAP-specific SQL generation prompt addendum (HANA dialect rules)		Not Started	02-Apr-2026			03-Apr-2026	SAP SQL addendum prompt	Blueprint Section 8 — double-quoted uppercase, LIMIT vs TOP, etc.
Implement agent factory (app/agent/factory.py) — create_agent with full middleware stack		Not Started	03-Apr-2026			06-Apr-2026	Agent factory module	Wire model, tools, middleware, checkpointer, response_format
Define structured output schema (AtcoGenieResponse) with response_format parameter		Not Started	06-Apr-2026			07-Apr-2026	Structured output schema	System, data, sources, metadata fields
Write unit tests for each middleware in isolation (mock state, mock runtime)		Not Started	07-Apr-2026			10-Apr-2026	Middleware test suite	Test security injection, tool filtering, usage tracking
Module 5: SQL Generation & Query Routing Engine		Not Started	27-Mar-2026			17-Apr-2026		
Implement ScopedSQLTool base class with execute(), _validate_tables(), _validate_no_mutations()		Not Started	27-Mar-2026			31-Mar-2026	ScopedSQLTool base class	Blueprint Section 9.1 — AST parsing, table allowlisting
Implement SQL AST validation using sqlparse (SELECT-only enforcement, subquery escape prevention)		Not Started	31-Mar-2026			03-Apr-2026	sql_guard.py	Parse SQL into AST; reject INSERT/UPDATE/DELETE/DROP; check nested queries
Implement mandatory WHERE clause injection (_inject_user_scope) via AST manipulation		Not Started	03-Apr-2026			06-Apr-2026	User scope injection logic	Force user filter as post-processing — last line of defense
Build pharma_sql_tool with Pharma-specific schema, T-SQL dialect, and scoping (rep_id/territory_rep_id)		Not Started	06-Apr-2026			08-Apr-2026	pharma_sql.py tool	Configure per-table scope columns for Pharma CRM
Build sap_sql_tool with SAP schema, HANA dialect, and scoping (ERNAM/USNAM)		Not Started	08-Apr-2026			11-Apr-2026	sap_sql.py tool	Include SAP SQL addendum; handle SAP authorization scope injection
Build thirdparty_sql_tool with Third-Party schema, T-SQL dialect, and scoping (evaluator_id/owner_id)		Not Started	11-Apr-2026			13-Apr-2026	thirdparty_sql.py tool	T-SQL dialect, Third-Party table allowlist
Implement per-system RBAC table allowlists (PHARMA_RBAC, SAP_RBAC, THIRDPARTY_RBAC)		Not Started	13-Apr-2026			15-Apr-2026	rbac.py — per-system access matrices	Blueprint Section 9.2 — viewer/manager/admin per system
Implement cross-system query handling (parallel tool calls for comparison queries)		Not Started	15-Apr-2026			16-Apr-2026	Cross-system synthesis logic	LangChain 1.0 ReAct loop parallel_tool_calls support
Implement ambiguity detection and clarification prompting in system prompt		Not Started	15-Apr-2026			16-Apr-2026	Disambiguation logic	Blueprint Section 7 — ask user when query maps to multiple systems
Write SQL generation tests per database (validate T-SQL output for MS SQL, HANA SQL for SAP)		Not Started	16-Apr-2026			17-Apr-2026	SQL generation test suite	Test dialect correctness, scope injection, table validation
Module 6: Security Implementation (Data Governance)		Not Started	06-Apr-2026			20-Apr-2026		
Define SecurityContext dataclass (immutable, per-request — user identity + per-system roles)		Not Started	06-Apr-2026			07-Apr-2026	context.py	Blueprint Section 9 — frozen dataclass
Implement SecurityContext hydration from JWT + Redis cached role profile		Not Started	07-Apr-2026			09-Apr-2026	Context hydration logic	Parse JWT claims, fetch CompositeRoleProfile from Redis, build SecurityContext
Implement per-system scope ID resolution (_resolve_scope_id per system)		Not Started	09-Apr-2026			10-Apr-2026	Scope ID resolver	Pharma=employee_id, SAP=ad_username, ThirdParty=employee_id
Implement row-level data isolation enforcement (mandatory WHERE injection validation)		Not Started	10-Apr-2026			12-Apr-2026	Row-level isolation verification	Verify every executed query contains user scope filter

Implement SQL injection prevention (parameterized execution, input sanitization)	Not Started	12-Apr-2026			14-Apr-2026	SQL injection guard	Never interpolate user values into SQL strings; use parameterized queries
Implement defense-in-depth audit logging (log every SQL query with user context)	Not Started	14-Apr-2026			16-Apr-2026	Security audit logger	Log: user_id, system, SQL executed, timestamp, result count
Implement no-access scenario handling (user-friendly message when system access is denied)	Not Started	16-Apr-2026			17-Apr-2026	Access denied handler	Blueprint Section 7 — inform user which systems they can access
Write security penetration tests (scope bypass attempts, SQL injection vectors, unauthorized tool access)	Not Started	17-Apr-2026			20-Apr-2026	Security test suite	Test every defense-in-depth layer
Module 7: Caching & Performance Optimization	Not Started	10-Apr-2026			24-Apr-2026		
Set up Redis connection with connection pooling (aioredis)	Not Started	10-Apr-2026			11-Apr-2026	Redis client configuration	Single Redis instance for cache, session, rate limiting
Implement exact-match query cache (QueryCache class with per-system TTLs)	Not Started	11-Apr-2026			14-Apr-2026	query_cache.py	Blueprint Section 10.1 — SHA256 hash key, scoped by user
Configure per-system cache TTLs (Pharma: 5min, SAP: 10min, ThirdParty: 15min)	Not Started	14-Apr-2026			15-Apr-2026	TTL configuration	Reference data has longer TTLs than transactional data
Implement cache invalidation strategy (TTL-based expiry, manual flush endpoint for admin)	Not Started	15-Apr-2026			16-Apr-2026	Cache invalidation logic	Admin endpoint to force-flush specific system caches
Implement SSE response streaming (FastAPI StreamingResponse with agent.astream)	Not Started	16-Apr-2026			19-Apr-2026	Streaming chat endpoint	Blueprint Section 10.2 — stream tokens + status updates
Implement status event streaming from middleware (e.g., Querying SAP HANA...)	Not Started	19-Apr-2026			20-Apr-2026	Status event emitter	Use stream_mode=['updates', 'custom'] for progressive feedback
Implement parallel tool execution support verification (LangChain 1.0 concurrent tool_calls)	Not Started	20-Apr-2026			21-Apr-2026	Parallel execution test	Blueprint Section 10.3 — verify cross-system queries execute concurrently
Performance benchmarking — measure p90 latency against target budgets	Not Started	21-Apr-2026			24-Apr-2026	Performance benchmark report	Blueprint Section 10.4 — target p90 < 3s common, < 5s complex
Module 8: Conversation Memory & State Management	Not Started	03-Apr-2026			17-Apr-2026		
Set up PostgreSQL instance for conversation checkpointing (PostgresSaver)	Not Started	03-Apr-2026			05-Apr-2026	PostgreSQL checkpointer DB	Separate from business DBs; lightweight instance
Configure PostgresSaver with create_agent checkpointer parameter	Not Started	05-Apr-2026			06-Apr-2026	Checkpointer integration	Blueprint Section 11 — auto-checkpoint on every turn
Implement Redis session store for active context (last_intent, last_target_system, resolved_entities)	Not Started	06-Apr-2026			08-Apr-2026	session.py	Blueprint Section 11 — TTL 30 min; hset with conversation_id key
Implement multi-turn context resolution (resolve references like 'it', 'that', 'show me more')	Not Started	08-Apr-2026			11-Apr-2026	Context resolution logic	Use last_target_system and resolved_entities from Redis session
Integrate SummarizationMiddleware with conversation checkpointer for long sessions	Not Started	11-Apr-2026			13-Apr-2026	Summarization + checkpointer integration	Ensure summarized history is persisted correctly
Implement conversation session management (create, resume, list, delete conversations)	Not Started	13-Apr-2026			15-Apr-2026	Session management API	Thread ID management for multi-turn support
Write multi-turn conversation tests (context carryover, entity resolution, summarization trigger)	Not Started	15-Apr-2026			17-Apr-2026	Conversation test suite	Test 10+ turn conversations with system switching
Module 9: Chat API & End-to-End Integration	Not Started	17-Apr-2026			01-May-2026		
Build POST /api/v1/chat endpoint with full request/response contract	Not Started	17-Apr-2026			20-Apr-2026	Chat endpoint	Blueprint Section 4 — message, conversation_id, stream params
Wire chat endpoint to LangChain agent with SecurityContext injection via config.context	Not Started	20-Apr-2026			22-Apr-2026	Agent invocation integration	Pass thread_id and security_context in config
Implement response formatting with source citation (system, SQL query, metadata)	Not Started	22-Apr-2026			24-Apr-2026	Response formatter	Include tokens_used, latency_ms, target_system, cache_hit
Implement error handling and graceful degradation (DB down, LLM timeout, invalid SQL)	Not Started	24-Apr-2026			26-Apr-2026	Error handling layer	Return user-friendly messages; log details for debugging
Implement conversation_id generation and management in chat flow	Not Started	26-Apr-2026			27-Apr-2026	Conversation ID logic	Auto-generate for new conversations; validate for existing
End-to-end integration testing — full flow from auth to chat to SQL execution to response	Not Started	27-Apr-2026			01-May-2026	E2E test suite	Test against all 3 databases; verify security context propagation
Module 10: Frontend Integration (React JS)	Not Started	17-Apr-2026			08-May-2026		
Define REST API contract between React frontend and FastAPI backend	Not Started	17-Apr-2026			19-Apr-2026	API specification document	OpenAPI/Swagger spec for /auth/login, /chat, /health
Implement JWT storage and Authorization header injection in React HTTP client	Not Started	19-Apr-2026			21-Apr-2026	Frontend auth integration	Store JWT securely; attach to every request
Build SSE event stream consumer in React for real-time response rendering	Not Started	21-Apr-2026			24-Apr-2026	SSE consumer component	Handle token events, status events, error events
Implement chat message UI with system source badges (Pharma/SAP/ThirdParty)	Not Started	24-Apr-2026			27-Apr-2026	Chat message rendering	Visual indicators for which database answered the query

Implement conversation history sidebar (list, resume, delete conversations)		Not Started	27-Apr-2026			30-Apr-2026	Conversation history UI	Persist conversation list; resume by thread_id
Implement login screen with AD credential input and error handling		Not Started	30-Apr-2026			02-May-2026	Login screen	Username/password form; display accessible systems on success
Implement loading states and progressive feedback (e.g., 'Querying SAP HANA...')		Not Started	02-May-2026			04-May-2026	Loading/status UI	Display middleware status events during query processing
Implement responsive layout and cross-browser compatibility testing		Not Started	04-May-2026			06-May-2026	Responsive UI	Test on Chrome, Edge, Firefox at minimum
Implement user feedback mechanism (thumbs up/down on responses)		Not Started	06-May-2026			08-May-2026	Feedback component	Send feedback to backend for quality monitoring dashboard
Module 11: Observability & Quality Assurance		Not Started	24-Apr-2026			08-May-2026		
Configure LangSmith/Langfuse integration with tracing callbacks		Not Started	24-Apr-2026			26-Apr-2026	Observability setup	Capture every agent step: routing decision, SQL generated, tool used
Implement routing accuracy logging (user_query, selected_tool, target_system, confidence)		Not Started	26-Apr-2026			28-Apr-2026	Routing accuracy tracker	Blueprint Section 7 — log classification decisions for review
Set up automated evaluation datasets for routing accuracy testing		Not Started	28-Apr-2026			01-May-2026	Eval datasets	Golden test set: query → expected target system mapping
Set up automated evaluation datasets for SQL quality testing		Not Started	01-May-2026			04-May-2026	SQL quality evals	Golden test set: query → expected SQL output per dialect
Implement quality monitoring dashboard (routing accuracy, SQL errors, user feedback)		Not Started	04-May-2026			06-May-2026	Monitoring dashboard	Track misrouted queries, failed SQL, negative feedback
Document weekly quality review process (review misrouted queries, refine system prompt)		Not Started	06-May-2026			07-May-2026	QA process documentation	Blueprint Section 7 — refine routing examples based on production traces
Build user feedback pipeline (thumbs up/down → quality monitoring ingestion)		Not Started	07-May-2026			08-May-2026	Feedback pipeline	Aggregate feedback for systematic improvement
Module 12: Deployment & Operations		Not Started	01-May-2026			15-May-2026		
Create production Dockerfile with multi-stage build (slim Python image)		Not Started	01-May-2026			03-May-2026	Dockerfile	Include ODBC drivers, hdbcli dependencies
Create production docker-compose.yml (FastAPI, Redis, PostgreSQL checkpointer)		Not Started	03-May-2026			05-May-2026	docker-compose.production.yml	Blueprint Section 12 — environment variable configuration
Implement Windows on-premises deployment scripts (NSSM for service management)		Not Started	05-May-2026			08-May-2026	Windows deployment scripts	PowerShell scripts for service install/start/stop
Configure HTTPS/TLS termination and certificate management		Not Started	08-May-2026			10-May-2026	TLS configuration	IIS reverse proxy or direct TLS on FastAPI
Create environment-specific configuration templates (dev, staging, production)		Not Started	10-May-2026			11-May-2026	Config templates	.env.dev, .env.staging, .env.production
Implement database migration scripts for checkpointer PostgreSQL schema		Not Started	11-May-2026			12-May-2026	Migration scripts	Alembic or manual SQL scripts for checkpointer tables
Create operational runbook (startup procedure, health checks, troubleshooting, cache flush)		Not Started	12-May-2026			14-May-2026	Operational runbook	Step-by-step guide for operations team
Perform production readiness review (security audit, performance validation, operational sign-off)		Not Started	14-May-2026			15-May-2026	Production readiness checklist	Final review before go-live